Reproducibility README for "Covariate-Assisted Bayesian Graph Learning for Heterogeneous Data"

1. Introduction

The supplements of reproducibility (under folder **"PxG_code"**) include four subfolders: **"Code"**, **"Data"**, **"RData"**, **"RMarkdown"**.

- Subfolder **"Code"** contains R and C++ scripts to reproduce all three simulation studies in section 4 and real data analysis of breast cancer in section 5. It also includes R package and C++ scripts of methods for comparison.

- Subfolder **"Data"** contains the breast cancer data from The Cancer Genome Atlas (TCGA, https://tcga-data.nci.nih.gov/tcga/)

- Subfolder **"RData"** contains several intermediary results from simulation studies and real data analysis. To save computational time, they can be loaded into R before generating any plots.

- Subfolder **"RMarkdown"** contains four PDF files generated by R Markdown. They provide the workflow for reproducing all simulation studies and real data analysis under the proposed PxG model.

Please see the following sections for detailed descriptions for the four subfolders listed above and instructions on how to reproduce simulation studies and real data analysis in this paper.

2. Data

Subfolder **"Data"** contains the breast cancer data from The Cancer Genome Atlas Program (TCGA, https://tcga-data.nci.nih.gov/tcga/). The data are downloaded using R package **"TCGA-Assembler"** (https://github.com/compgenome365/TCGA-Assembler-2). **"BRCA_RNA.RData"** contains mRNA data of 883 samples corresponding to three genes: ERBB2, ESR1 and PGR. **"BRCA_RPPA.RData"** contains protein data of 883 samples corresponding to 15 genes: PTEN, PIK3CA, BAD, NFKB1, TSC1, IRS1, EIF4EBP1, EGFR, AKT1, TSC2, MTOR, DVL3, ERBB2, ESR1, PGR.

3. Code: R and C++ Scripts

Subfolder **"Code"** contains all R and C++ scripts to reproduce figures and numbers reported in the paper. There are 8 files under this folder (5 R files, 1 cpp file, 1 zip file and 1 R package). Below is a list of description for each file.

- **"PxG-GW.R":** This R script contains all functions for the proposed PxG model with Gaussian-G-Wishart specification in section 3.2.1. It is used for simulation studies in section 4.1 and 4.2. It needs to be loaded in **"Example1.R"** and **"Example2.R"**.

- **"PxG-PL.cpp":** This C++ script contains all functions for the proposed PxG model with pseudo-likelihood specification in section 3.2.2. It is used for simulation studies in section 4.3 and real data analysis in section 5. It needs to be loaded and complied in **"Example3.R"** and **"RealDataAnalysis.R"**.

- **"Example1.R":** This R script reproduces simulation results in section 4.1 "Example 1: piecewise linear precision matrix" including values of DIC and MSE for different models, Figure 3: posterior edge inclusion probability, and Figure 4: posterior estimates of partial correlations in the paper.

- **"Example2.R":** This R script reproduces simulation results in section 4.2 "Example 2: linear precision matrix with constant graph" including values of MSE for the PxG model, Figure 5: posterior estimates of partial correlations under the PxG model, and Figure 6: posterior estimates of edge inclusion probability (BM13) or partial correlation (mSSL).

- **"Example3.R":** This R script reproduces simulation results in section 4.3 "Example 3: piecewise constant precision matrix" including values of DIC for different models, Figure 7: posterior clustering results for all three models, Figure 8: posterior edge inclusion probability for graphs in both clusters under PxG, and Figure 9: posterior results regarding graph estimates under BM13, mSSL and JRNS.

- **"RealDataAnalysis.R":** This R script reproduces real data analysis results in section 5 "Real Data Analysis: Breast Cancer Data" including values of total sum of squares of covariates for different models, Figure 10: posterior clustering results for the PxG model, Figure 11: posterior clustering results for the graph-only model, and Figure 12: posterior edge inclusion probability for each cluster under the PxG model.

- **"LSGraph_0.1.tar.gz":** This R package is for the BM13 model (Bhadra and Mallick, 2013; Niu et al., 2020). It is used in section 4.2 (**"Example2.R"**) and 4.3 (**"Example3.R"**) for model comparison.

- **"JRNS.zip":** This zip file contains three C++ scripts for the JRNS model (Samanta et al., 2022). It is used in section 4.3 (**"Example3.R"**) for model comparison.

There are three major functions in **'PxG-GW.R'**. For all three functions, the Gaussian-G-Wishart model is used for likelihood function and graphical prior.

- The proposed PxG model
  PxG_GW = function(Y, X, b, D, alphaG, mu0, sigsq0, b1, b2, M, N, burnin, nmcmc, z0, B)

- The covariate-only model
  CovOnly_GW = function(Y, X, b, D, alphaG, mu0, sigsq0, b1, b2, M, N, burnin, nmcmc, z0, B)

- The graph-only model
  GraphOnly_GW = function(Y, X, b, D, alphaG, mu0, sigsq0, b1, b2, M, N, burnin, nmcmc, z0, B)

➢ Argument list
  - **Y:** response data or graphical data (n by q)
  - **X:** covariate data (n by p)
  - **b:** degree of freedom for G-Wishart distribution
  - **D:** positive definite matrix for G-Wishart distribution (q by q)
  - **alphaG:** prior edge inclusion probability
  - **mu0:** mean of normal priors in the auxiliary model, see section 3.1
  - **sigsq0:** variance of priors in the auxiliary model, see section 3.1
  - **b1:** shape parameter of inverse gamma prior in the auxiliary model, see section 3.1
  - **b2:** rate parameter of inverse gamma prior in the auxiliary model, see section 3.1
  - **M:** scaling parameter in the stick-breaking process
  - **N:** maximum number of clusters allowed
  - **burnin:** number of burn-in samples
  - **nmcmc:** number of MCMC samples
  - **z0:** initial cluster allocation
  - **B:** number of graph updates in each iteration

➢ Output list
  - **Label:** posterior estimates of cluster labels using the least squares clustering method (Dahl, 2006)
  - **Gsum:** posterior edge inclusion probability of graph for each sample
  - **Osum:** posterior precision matrix for each sample
  - **avgdist:** average least square distance to the mean for all MCMC samples of cluster labels (Dahl, 2006)
  - **DIC:** DIC value
  - **LogTotLike:** log total likelihood under the MCMC sample (the one in "**Label**") selected by the least squares clustering method (Dahl, 2006)
  - **VarTotLike:** variance of log total likelihood under all MCMC samples of cluster labels

There are three major functions in **'PxG-PL.cpp'**. For all three functions, the pseudo-likelihood model is used for likelihood function and graphical prior.

- The proposed PxG model
  PxG_PL(arma::mat YData, arma::mat XData,
  double spike, double slab, double a1, double a2, double alphaG,
  arma::vec mu0, double sigsq0, double b1, double b2,
  double M, unsigned int N,
  unsigned int burnin, unsigned int nmcmc, arma::uvec z0, unsigned int B)

- The covariate-only model
  CovOnly_PL(arma::mat YData, arma::mat XData,
  double spike, double slab, double a1, double a2, double alphaG,
  arma::vec mu0, double sigsq0, double b1, double b2,
  double M, unsigned int N,
  unsigned int burnin, unsigned int nmcmc, arma::uvec z0)

- The graph-only model
  GraphOnly_PL(arma::mat YData, arma::mat XData,
  double spike, double slab, double a1, double a2, double alphaG,
  arma::vec mu0, double sigsq0, double b1, double b2,
  double M, unsigned int N,
  unsigned int burnin, unsigned int nmcmc, arma::uvec z0, unsigned int B)

- Argument list
  - **YData:** response data or graphical data (n by q)
  - **XData:** covariate data (n by p)
  - **spike:** "spike" variance parameter of the spike-and-slab prior, see section 3.2.2
  - **slab:** "slab" variance parameter of the spike-and-slab prior, see section 3.2.2
  - **a1:** shape parameter of inverse gamma prior for the variance parameter of the spike-and-slab prior, see section 3.2.2
  - **a2:** rate parameter of inverse gamma prior for the variance parameter of the spike-and-slab prior, see section 3.2.2
  - **alphaG:** prior edge inclusion probability
  - **mu0:** mean of normal priors in the auxiliary model, see section 3.1
  - **sigsq0:** variance of priors in the auxiliary model, see section 3.1
  - **b1:** shape parameter of inverse gamma prior in the auxiliary model, see section 3.1
  - **b2:** rate parameter of inverse gamma prior in the auxiliary model, see section 3.1
  - **M:** scaling parameter in the stick-breaking process
  - **N:** maximum number of clusters allowed
  - **burnin:** number of burn-in samples
  - **nmcmc:** number of MCMC samples
  - **z0:** initial cluster allocation
  - **B:** number of graph updates in each iteration

- Output list
  - **Label:** posterior estimates of cluster labels using the least squares clustering method (Dahl, 2006)
  - **DIC:** DIC value
  - **LogTotLike:** log total likelihood under the MCMC sample (the one in **"Label"**) selected by the least squares clustering method (Dahl, 2006)
  - **Var:** variance of log total likelihood under all MCMC samples of cluster labels
  - **Gsum:** 3-dimensional array (q-1 by q by n), graph selection results for each sample using pseudo-likelihood function

4. R Packages Required

- To run **"Example1.R"** for reproducing simulation results in section 4.1, please install R packages: `huge`, `invgamma`, `BDgraph`, `ggplot2`, `ggpubr`.

- To run **"Example2.R"** for reproducing simulation results in section 4.2, please install R packages: `invgamma`, `BDgraph`, `devtools`, `mSSL` (from GitHub), `SSLASSO`, `MASS`, `LSGraph` (included in the **"Code"** subfolder), `ggplot2`, `ggpubr`.

- To run **"Example3.R"** for reproducing simulation results in section 4.3, please install R packages: `invgamma`, `BDgraph`, `Rcpp`, `RcppArmadillo`, `devtools`, `mSSL` (from GitHub), `SSLASSO`, `MASS`, `LSGraph` (included in the **"Code"** subfolder), `gglasso`, `glasso`, `glmnet`, `mvtnorm`, `rms`, `dlm`, `ggplot2`, `ggpubr`.

- To run **"RealDataAnalysis.R"** for reproducing real data analysis results in section 5, please install R packages: `Rcpp`, `RcppArmadillo`, `ggplot2`, `ggpubr`.

5. R Markdown Files

We include four R Markdown files and their PDF files in the **"RMarkdown"** subfolder. They provide detailed workflow for reproducing all simulation studies and real data analysis under only the proposed PxG model (without comparison with other models).

- **"Example1.Rmd" and "Example1.pdf":** reproducing simulation results in section 4.1 under the PxG model
- **"Example2.Rmd" and "Example2.pdf":** reproducing simulation results in section 4.2 under the PxG model
- **"Example3.Rmd" and "Example3.pdf":** reproducing simulation results in section 4.3 under the PxG model

- **"RealDataAnalysis.Rmd" and "RealDataAnalysis.pdf":** reproducing real data analysis in section 5 under the PxG model

6. Intermediary Results in RData

Subfolder **"RData"** contains several intermediary results from simulation studies and real data analysis. To save computational time, they can be loaded into R before generating any plots. See the corresponding R scripts for more details.

- From **"Example1.R":**
  - **"Example1A.RData":** simulation results under the PxG model in section 4.1
  - **"Example1B.RData":** simulation results under the covariate-only model in section 4.1
  - **"Example1C.RData":** simulation results under the graph-only model in section 4.1
  - **"klasso.RData":** simulation results under the kernel LASSO approach in section 4.1

- From **"Example2.R":**
  - **"Example2.RData":** simulation results under the PxG model in section 4.2

- From **"Example3.R":**
  - **"Example3A.RData":** simulation results under the PxG model in section 4.3
  - **"Example3AG.RData":** simulation results of graph estimate for each cluster under the PxG model in section 4.3
  - **"Example3B.RData":** simulation results under the covariate-only model in section 4.3
  - **"Example3C.RData":** simulation results under the graph-only model in section 4.3

- From **"RealDataAnalysis.R":**
  - **"RealDataAnalysisA.RData":** clustering results of fitting the breast cancer data with the PxG model
  - **"RealDataAnalysisAG.RData":** results of posterior graph estimate for each cluster under the PxG model
  - **"RealDataAnalysisC.RData":** clustering results of fitting breast cancer data with the graph-only model

7. Instructions For Use

Before running any R scripts, install all required R packages listed above. To reproduce all simulation studies and real data analysis using R scripts in the subfolder **"Code":**

- To reproduce all simulation results and figures in section 4.1, directly run **"Example1.R"** in the subfolder **"Code"**.

- To reproduce all simulation results and figures in section 4.2, first install **mSSL** package (see line 152-154 in **"Example2.R"** for instructions) and install **LSGraph** package (see line 195-196 in **"Example2.R"** for instructions), then run **"Example2.R"** in the subfolder **"Code"**.

- To reproduce all simulation results and figures in section 4.3, first install **mSSL** package (see line 221-223 in **"Example3.R"** for instructions) and install **LSGraph** package (see line 268-269 in **"Example3.R"** for instructions), then unzip **"JRNS.zip"** for three C++ scripts (**"JRNS.cpp"**, **"StepB.cpp"**, **"StepOmega.cpp"**) required for the **JRNS** model. Finally, run **"Example3.R"** in the subfolder **"Code"**.

- To reproduce all results and figures in section 5, first load breast cancer data (**"BRCA_RNA.RData"**, **"BRCA_RPPA.RData"**) from the subfolder **"Data"**, then run **"RealDataAnalysis.R"** in the subfolder **"Code"**.

Before running any R Markdown files, install all required R packages listed above. To reproduce simulation studies and real data analysis under ONLY the PxG model using R Markdown files in the subfolder **"RMarkdown"**, first copy **"PxG_GW.R"** and **"PxG_PL.cpp"** from the subfolder **"Code"** to the subfolder **"RMarkdown"**, copy **"BRCA_RNA.RData"** and **"BRCA_RPPA.RData"** from the subfolder **"Data"** to the subfolder **"RMarkdown"**, then:

- To reproduce simulation results and figures under only the PxG model in section 4.1, directly run **"Example1.Rmd"** in the subfolder **"RMarkdown"**.

- To reproduce simulation results and figures under only the PxG model in section 4.2, directly run **"Example2.Rmd"** in the subfolder **"RMarkdown"**.

- To reproduce simulation results and figures under only the PxG model in section 4.3, directly run **"Example3.Rmd"** in the subfolder **"RMarkdown"**.

- To reproduce results and figures under only the PxG model in section 5, directly run **"RealDataAnalysis.Rmd"** in the subfolder **"RMarkdown"**.